

EPUSTAKA MUNSHI: SMART LIBRARY SYSTEM WITH OCR LEDGER DIGITIZATION

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Abstract - Secondary-school libraries that still depend on handwritten ledgers face persistent problems in catalog search, book-status tracking, borrowing control, and preservation of historical acquisition records. This paper presents ePustaka Munshi, a web-based smart library system developed for SMK Abdullah Munshi to modernize a paper-based library workflow. The system integrates book and copy cataloguing, student and member management, role-based staff access, barcode-based checkout and return, bilingual user interfaces, and an OCR-assisted workflow for digitizing handwritten Buku Induk records. A React frontend, Flask REST API, and Supabase PostgreSQL database form the main three-tier architecture, while Anthropic Claude Vision is used as the primary handwritten-ledger extraction engine with Tesseract retained as a traditional OCR baseline and page-orientation aid. The system was developed using Agile Scrum across four functional sprints: user and role management, catalog and inventory, ledger digitization, and circulation. Testing covered functional validation, integration checks, OCR review correction, and user acceptance feedback from the school library context. The completed system reduces dependence on fragile paper records, provides a searchable digital catalogue, supports faster barcode circulation, and preserves historical ledger data through a human-verified OCR process.

Keywords - Smart library system; OCR ledger digitization; handwritten document transcription; barcode circulation; React; Flask; Supabase.

I. INTRODUCTION

Library resource management is a practical requirement for supporting reading culture, academic reference use, and day-to-day school administration. At SMK Abdullah Munshi in Penang, Malaysia, library operations had long depended on physical ledgers, manual borrowing logs, and simple spreadsheet records. The school library manages more than 7,000 books, yet important fields such as titles, authors, publishers, Dewey Decimal Classification, acquisition numbers, and loan statuses were fragmented across physical and digital records.

The manual process created three connected problems. First, the physical ledger was difficult to search and vulnerable to deterioration. Second, borrowing and returning required repeated manual writing, especially during peak school hours. Third, previous digitization attempts did not provide validation, duplication checking, real-time availability, or role-based permissions. Similar school-library modernization challenges have been reported in web-based library system studies, where paper ledgers restrict searchability and increase record-loss risk [1], [6].

This project therefore developed ePustaka Munshi, a web-based library management system with OCR ledger digitization. The goal is not merely to replace a paper ledger with a database, but to create a workflow that fits the real operating conditions of a Malaysian secondary-school library: teachers supervise, library prefects help at the counter, students need simple access to catalogue and loan information, and old handwritten records must be preserved before they degrade.

II. BACKGROUND AND RELATED WORK

Digital transformation in school libraries is increasingly linked to collection visibility, administrative efficiency, and equitable access to learning resources. Studies on digital library management and digital collection strategies emphasize that school libraries benefit when metadata, circulation, and user records are centralized rather than kept in isolated physical files [1], [11].

Existing library platforms were evaluated against the needs of SMK Abdullah Munshi. KOHA provides mature integrated library functionality, but its server administration and cataloguing complexity make it heavy for a small secondary-school deployment. Legacy desktop systems such as e-Kutub Khanah match older Malaysian school contexts, but are vulnerable to device dependency, compatibility issues, and data-loss risk. Spreadsheet-based approaches are simple, but they do not provide circulation automation, permission control, or OCR-assisted migration of handwritten records.